

UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS



Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

METHOD FIT MATRIX

DELIVERABLE No. 2

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2026 - II

Scoped research question: How can a human-in-the-loop multimodal AAC prototype in Spanish be designed and rigorously evaluated to support independent functional communication in one clearly defined group of minimally speaking or nonspeaking autistic children requiring very substantial support, within one structured therapeutic context?

Criterion / Weight	Pure Experiment	Pure Case Study	Mixed Methods	Design Science Research
E: Epistemological Fit 30%	3/5 — Can test effects, but does not govern artifact design.	2/5 — Strong contextual depth, weak fit with artifact-centered knowledge claim.	4/5 — Integrates evidence, but is broader than the core contribution.	5/5 — Natural fit: knowledge is generated through building and evaluating the artifact.
D: Data Feasibility 20%	2/5 — Randomized human data are difficult to obtain ethically and practically.	3/5 — Deep access and permissions are possible but resource-intensive.	3/5 — Requires both repeated quantitative data and qualitative recruitment.	4/5 — Prototype logs, repeated user observations, and focused feedback are obtainable.
F: Feasibility 15%	2/5 — High ethics, recruitment, randomization, and control burden within 3 years.	3/5 — Feasible, but prolonged immersion and permissions add time.	3/5 — Feasible only if tightly scoped; full mixed-methods study risks overextension.	4/5 — Feasible if limited to one prototype, one user group, one context, 2–3 outcomes.
C: Contribution 20%	3/5 — Produces effect evidence but not the intended design knowledge.	2/5 — Explains context but does not deliver the intended technological artifact.	4/5 — Produces integrated evidence, but artifact/design knowledge is not central.	5/5 — Delivers artifact + rigorous evaluation + transferable design principles.
V: Venue Fit 15%	4/5 — Publishable in evaluation-focused venues if causal design is defensible.	3/5 — Possible in HCI/accessibility venues, but weaker match to technical contribution.	4/5 — Well accepted in HCI/accessibility research, but broader study burden.	5/5 — Strong fit with ACM ASSETS, TACCESS, and AAC-oriented journals.
Weighted Total $E \times .30 + D \times .20 + F \times .15 + C \times .20 + V \times .15$	2.80/5 (56/100)	2.50/5 (50/100)	3.65/5 (73/100)	4.65/5 (93/100)

Decision: Design Science Research is selected as the primary method. Pure Experiment and Pure Case Study are eliminated by the ≤ 2 gate. Mixed Methods remains a viable evaluation strategy, but not the governing method. To remain feasible within three years, the DSR study is restricted to one functional prototype, one user subgroup, one structured context, and 2–3 primary outcomes. A narrowly scoped single-case comparative evaluation and focused stakeholder feedback may be embedded inside the DSR evaluation cycle. Color legend: Red indicates weak methodological fit (≤ 2); amber, moderate fit (3); green, strong methodological fit (≥ 4); blue, structural.

Gate rule: any candidate scoring ≤ 2 on a critical criterion is removed. **Simulation:** not shortlisted as the primary method because the central claim concerns real-user utility; it may still be used for pre-deployment stress testing of failure scenarios.

Data sources for the selected DSR plan: prototype interaction logs; coded independent functional communication acts; communicative diversity; prompting/assistance level; rejected/modified AI suggestions; focused caregiver and specialist feedback. Ethics approval, caregiver consent, child assent/withdrawal procedures, data minimization, and secure/on-device processing where feasible are prerequisites.